

AgriScan 360

Automated Multi-Modal Non-Destructive Produce Quality, Internal Defect, and Freshness Inspection System

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Project Focus: Embedded Systems, Multi-Modal Sensor Fusion, Agricultural Robotics, Edge AI

Target Commodities: Tomato (*Solanum lycopersicum*), Apple (*Malus domestica*), Eggplant (*Solanum melongena*)

Dataset Repository: huggingface.co/datasets/SHAFI6196UIU/agrican360-dataset

1. Abstract & Introduction

Abstract: Post-harvest agricultural loss represents a monumental economic crisis in developing nations such as Bangladesh, where 25% to 40% of fresh fruits and vegetables spoil prior to retail distribution. Existing sorting methods in regional supply chains remain archaic: they rely either on subjective surface-level human visual sorting (which fails to detect internal rot, hollow heart, or latent fungal pathogens) or on destructive penetrometer tests (which destroy market-ready produce). **AgriScan 360** resolves this trade-off by engineering an automated, multi-physics, non-destructive freshness inspection ecosystem. The device incorporates a gravity pipe feed chute actuated by an SG90 micro-servo motor that deposits produce into a 27-liter sealed dark chamber onto an 8-stop NEMA 17 stepper motor turntable (45° indexing). A Raspberry Pi 5 coordinates simultaneous multi-modal sensing: continuous Volatile Organic Compound (VOC) headspace sniffing via a Bosch BME688 MOX gas sensor runs concurrently with 16 synchronized optical captures (8 White-light RGB and 8 365 nm UV-A autofluorescence frames). Telemetry and dual-spectral imagery are streamed to a local host server running FastAPI and SQLite. A multi-modal decision engine fuses visual necrotic browning, fungal excitation, and dynamic gas resistance kinetics (resistance drop, ratio, and decay slope) into a binary classification (**HEALTHY** vs. **ROTTEN**). Results are displayed locally on an SSD1306 OLED screen and on an interactive web dashboard. Following the diagnostic cycle, the scan terminates safely without automated sorting, allowing the human operator to manually pick up the inspected item.

Introduction: Fruits and vegetables continue active metabolic respiration, transpiration, and volatile organic acid synthesis post-harvest. Traditional single-angle computer vision fails to detect early rot when decay occurs on the opposite side or internally near the seed cavity. Furthermore, latent fungal pathogens like *Botrytis cinerea* and *Penicillium expansum* remain visually undetectable to the human eye for days before surface mycelium appears. AgriScan 360 addresses these physical failure modes through three synchronized sensing pillars: (1) 360° Visible RGB Reflectance across 8 circumferential angles; (2) 365 nm UV-A Autofluorescence Excitation for latent fungal colonies; and (3) Chemical Headspace E-Nose Kinetics tracking dynamic VOC emissions inside a sealed 27L enclosure.

2. Project Objectives

- **Comprehensive 360° Spatial Assessment:** Eliminate optical blind spots by rotating produce across 8 discrete 45° stops using a precision stepper-driven turntable.
- **Simultaneous Optical & Chemical Acquisition:** Overlap dual-spectrum optical imaging (White RGB + 365 nm UV-A) with continuous chemical gas sniffing, reducing total testing time from ~5 minutes to ~3 minutes.
- **Automated Gravity Feed Mechanism:** Implement a pipe-loading flap door driven by an SG90 micro-servo (180° open, 90° closed) that deposits produce directly onto the turntable center.
- **Calibrated Vision ROI Crop:** Isolate the fruit inside the concentric turntable ring (X: 25%–76%, Y: 41%–100%), cropping out background chamber walls and wiring for clean AI inference.
- **Robust Binary Freshness Classification:** Classify produce strictly into HEALTHY or ROTTEN categories, incorporating an automatic gas override rule for latent internal decay.
- **Affordable Hardware Budget:** Deliver the complete system under 25,000 BDT using commercial-off-the-shelf components sourced from Bangladesh suppliers (e.g., RoboticsBD.com).

3. Key System Features

- **Inspect-and-Halt Workflow:** Operates on an inspect-and-halt protocol without mechanical sorters or diverters. Produce is loaded via pipe, inspected across all modalities, and manually retrieved by the operator.
- **Dual Solid-State MOSFET Switching:** Two independent IRLZ44N logic-level MOSFETs switch the 12V White LED array and 365 nm UV-A LED array via 3.3V GPIO commands.
- **Dynamic Gas Kinetics Engine:** Continuously extracts baseline resistance, minimum resistance, net drop, drop ratio, and decay velocity slope using linear regression.
- **Internal Decay Gas Override:** If produce appears visually clean but headspace VOC drop exceeds thresholds (ratio $\geq 15\%$ or drop $\geq 5.5 \text{ k}\Omega$ for Tomato), the system automatically overrides vision and marks the produce ROTTEN.
- **Dual HMI Display:** Features an onboard SSD1306 OLED display for standalone operation and a real-time web dashboard broadcasting 16-frame image carousels and gas curves over WebSocket.
- **Live Telemetry Sync:** Automatically exports all completed scans and ground-truth labels from SQLite (agriscan360.db) to CSV (bme688_telemetry_dataset.csv) for ongoing machine learning model retraining.

4. Hardware Components Specification

Component	Model / Specs	Role in AgriScan 360
Raspberry Pi 5 (4GB)	Broadcom BCM2712 Quad Cortex-A76 @ 2.4GHz	Edge orchestrator, I2C master, GPIO controller, HTTP client
Pi Camera Module 2	Sony IMX219 8.08MP, 1080p capture	Dual-spectrum RGB and UV-A optical imaging
BME688 MOX Sensor	Bosch Sensortec, I2C address 0x77	Headspace VOC resistance, temperature, humidity, pressure
NEMA 17 Stepper	17HS4401, 1.8° step, 40 N·cm torque	8-stop 360° turntable rotation
A4988 Motor Driver	Allegro Bipolar Microstepping Driver	Current-controlled drive for NEMA 17 stepper
SG90 Micro Servo	TowerPro 9g, 50 Hz PWM, 1.8 kg·cm	Pipe feed chute entry door flap (180° open / 90° closed)
SSD1306 OLED (0.96")	128×64 dot-matrix, I2C address 0x3C	Real-time local status and classification display
IRLZ44N MOSFETs (x2)	Logic-level N-Channel, $V_{gs(th)} \leq 2.0V$	3.3V logic switching for 12V White and UV-A LEDs
365 nm UV-A LED Array	High-radiance 365 nm UV emitters	Latent fungal mycelium autofluorescence excitation
White LED Array	High-CRI diffused 12V LED array	Glare-free visible surface reflectance lighting
12V 2A DC SMPS	Regulated 12V 2A Power Adapter	Motor and LED illumination power rail
3.7V 18650 Battery	Li-ion cell with 5V boost circuit	Portable / backup logic supply
Test Chamber	27-liter sealed dark box	Sealed atmosphere for headspace gas accumulation

5. Cost Analysis in BDT (RoboticsBD Benchmark)

#	Component Name	Source Reference	Unit Price (BDT)	Total (BDT)
1	Raspberry Pi 5 (4GB RAM)	RoboticsBD / TechshopBD	15,500	15,500
2	Pi Camera Module 2 (IMX219)	RoboticsBD.com	2,450	2,450
3	NEMA 17 Stepper Motor (17HS4401)	RoboticsBD.com	1,250	1,250
4	A4988 Stepper Driver Module	RoboticsBD.com	140	140
5	TowerPro SG90 9g Micro Servo	RoboticsBD.com	140	140
6	Bosch BME688 Gas Sensor Module	RoboticsBD.com	1,850	1,850
7	SSD1306 0.96" I2C OLED Display	RoboticsBD.com	320	320
8	365 nm High-Power UV-A LED Array	RoboticsBD / Local Market	450	450
9	High-CRI White Diffused LED Strip	Local Market	220	220
10	Dual IRLZ44N MOSFET Board	Local Electronics Market	180	180
11	12V 2A DC Power Adapter (SMPS)	RoboticsBD.com	420	420
12	3.7V 18650 Battery + Holder	RoboticsBD.com	320	320
13	27L Chamber Box, Pipe, Acrylic Turntable	Local Fabrication	1,400	1,400
14	Jumpers, Perfboard, Fasteners	Local Electronics Market	350	350
TOTAL	Complete AgriScan 360 System	Full Multi-Modal Platform	—	24,990 BDT

Economic Viability: Industrial single-modal fruit sorters (Tomra, Compac, Aweta) range from \$35,000 to \$120,000 USD (>40,00,000 BDT). AgriScan 360 delivers multi-spectral optical and chemical E-Nose diagnostics for **under 25,000 BDT** (~\$210 USD)—a 99% cost reduction.

6. Circuit Diagram & Hardware Interconnections

Component Pin	Target Pin on Raspberry Pi 5	Signal Type	Electrical Characteristic
A4988 STEP	GPIO 17 (Physical Pin 11)	Digital Output	3.3V pulse (5 ms delay)
A4988 DIR	GPIO 27 (Physical Pin 13)	Digital Output	HIGH = Clockwise, LOW = CCW
A4988 ENABLE	GPIO 22 (Physical Pin 15)	Digital Output	Active-LOW: LOW = Motor ON, HIGH = Motor OFF
A4988 VMOT / GND	External 12V 2A DC Supply	Power Rail	12V DC with 100 μ F decoupling capacitor
A4988 VDD / GND	3.3V (Pin 1) & GND (Pin 6)	Logic Supply	3.3V logic reference
SG90 Servo PWM	GPIO 23 (Physical Pin 16)	Hardware PWM	50 Hz PWM (90° closed, 180° open)
SG90 VCC / GND	5V Rail (Pin 2) & GND (Pin 14)	Servo Power	5.0V DC operating supply
White MOSFET Gate	GPIO 18 (Physical Pin 12)	Digital Output	3.3V logic with 10 k Ω pulldown
UV-A MOSFET Gate	GPIO 24 (Physical Pin 18)	Digital Output	3.3V logic with 10 k Ω pulldown
BME688 SDA	GPIO 2 (Physical Pin 3)	I2C Data	Shared I2C bus, address 0x77
BME688 SCL	GPIO 3 (Physical Pin 5)	I2C Clock	Shared I2C bus clock
SSD1306 SDA	GPIO 2 (Physical Pin 3)	I2C Data	Shared I2C bus, address 0x3C
SSD1306 SCL	GPIO 3 (Physical Pin 5)	I2C Clock	Shared I2C bus clock
Pi Camera Module 2	CAM0 / CAM1 Port	MIPI CSI-2	15-pin high-speed serial ribbon

7. Working Principle & Implementation Details

Step A — Clean-Air Baseline Sniffing: The empty 27L chamber is closed. The BME688 pre-heats and sniffs for 180 seconds. The first 120 seconds are discarded for thermal stabilization; the final 5 seconds establish the reference baseline resistance R_{base} .

Step B — Pipe Drop & Auto-Detection: The operator drops produce into the top pipe. The SG90 servo opens to 180° for 2 seconds, releasing the fruit onto the turntable, and closes to 90°. The lid is secured. The camera captures a test image with ROI cropping (X: 25%–76%, Y: 41%–100%) and classifies the fruit type (Tomato, Apple, or Eggplant) via HSV color space analysis.

Step C — Simultaneous Sniffing & 8-Stop Scan: Continuous BME688 sniffing launches on a background thread. Simultaneously, the NEMA 17 motor steps through 8 angular positions (45° each). At each stop: (1) White LED activates → captures RGB frame → turns off; (2) UV-A LED activates → captures UV fluorescence frame → turns off; (3) Turntable advances 25 steps (45°). All 16 photos finish in ~100 seconds. If incubation time remains, the system counts down the remaining seconds (skippable via Ctrl+C).

Step D — Decision Fusion & Result Display: The Pi stops gas sniffing, compiles net drop, drop ratio, and decay slope, and uploads all 16 frames + gas telemetry to the FastAPI server. The AI classifies the sample: Fused Score = $0.40 \cdot P_1 + 0.35 \cdot P_2 + 0.25 \cdot P_3$. If Fused Score ≥ 0.40 or $P_3 \geq 0.70$ (gas override), status is ROTTEN; otherwise HEALTHY.

Step E — Manual Pickup (No Sorting Mechanism): The status is shown on the OLED display and web UI. Motor coils de-energize. **The machine halts completely.** The operator manually opens the lid, removes the fruit, and sorts it based on the displayed result.

8. Experimental Results & Practical Applications

Scan ID	Produce	Ground Truth	R_base	R_post	Delta R	Ratio %	AI Result	Confidence
08	Tomato	HEALTHY	270.66 kΩ	267.02 kΩ	3.64 kΩ	2.26%	HEALTHY	88.2%
11	Tomato	ROTTEN	263.15 kΩ	212.98 kΩ	50.17 kΩ	19.65%	ROTTEN	94.0%
13	Tomato	ROTTEN	259.66 kΩ	251.72 kΩ	7.94 kΩ	4.21%	ROTTEN	78.5%
14	Tomato	HEALTHY	268.38 kΩ	262.99 kΩ	5.39 kΩ	4.47%	HEALTHY	87.5%

Applications: (1) Wholesale intake checking to isolate batches with latent fungal rot before transit; (2) Cold storage pre-entry screening to avoid storing fruits that emit ethylene and spoil adjacent produce; (3) Non-destructive quality verification of premium retail produce.

9. Social Impact & Future Scope

Social Impact in Bangladesh: Reducing post-harvest waste directly preserves marketable volumes of high-value crops (tomatoes, eggplants) without requiring additional arable land or fertilizer. It also provides smallholder farmers with an objective, low-cost freshness evaluation to prevent unfair price discounts from middlemen, while protecting public health by identifying fungal mycotoxins before produce reaches consumer markets.

Future Scope: (1) *Automated Diverter Arm:* Add a dual-bin servo diverter to sort items into HEALTHY and REJECT bins automatically, eliminating manual pickup; (2) *Conveyor Integration:* Replace the turntable with an indexed roller conveyor to increase throughput from ~20 to >600 items/hour; (3) *Vis-NIR Spectroscopy:* Integrate an AS7265x multi-spectral sensor (410–940 nm) for non-destructive sugar content (°Brix) and internal moisture measurement; (4) *Standalone Edge NPU:* Quantize models to INT8 to run entirely on the Raspberry Pi 5 using a Hailo-8L NPU accelerator, removing the need for an external laptop server.

10. Conclusion & References

Conclusion: AgriScan 360 demonstrates that multi-modal sensing combining 360° optical imaging, UV-A autofluorescence, and dynamic metal-oxide headspace kinetics provides reliable, non-destructive produce freshness evaluation. Running 16-photo image capture simultaneously with continuous BME688 sniffing delivers complete inspections within ~3 minutes. With an inspect-and-halt protocol and manual retrieval, the system achieves commercial-grade evaluation at a bill-of-materials cost under 25,000 BDT, providing a practical, accessible solution for post-harvest quality assessment in developing agricultural economies.

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